
Capability-Level Diversity in Multi-Agent LLM Systems: An Empirical Study on Adversarial Collective Reasoning

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Abstract

Multi-agent LLM systems increasingly power critical reasoning tasks, yet most deployments use homogeneous model ensembles that may share correlated failure modes. Classical ensemble theory predicts that agent diversity reduces correlated errors, but whether *capability-level* diversity within a single model family provides measurable benefits remains an open empirical question. We study this question by constructing ADVREASON-30, a 30-question adversarial reasoning benchmark spanning five cognitive trap categories—misleading math, causal traps, logical deception, numerical tricks, and framing effects—and evaluating five experimental conditions using real API calls to CLAUDE-HAIKU-4-5 and CLAUDE-SONNET-4-6 models. Our conditions cover a diversity continuum from single agents, to homogeneous three-agent ensembles using majority voting, to heterogeneous ensembles mixing small and large models, and a structured debate protocol. We find that both individual models achieve high accuracy (93.3%) on our benchmark, leaving little room for diversity-based improvement. Error correlation between the cross-capability CLAUDE-HAIKU-4-5–CLAUDE-SONNET-4-6 pair ($r=0.464$) is measurably lower than within-capability pairs ($r=0.695$), partially supporting the diversity hypothesis. However, heterogeneous ensembles match—rather than exceed—homogeneous ensembles at 96.7% accuracy ($p=1.000$, McNemar), and the debate protocol slightly degrades performance due to sycophantic revision. The dominant shared failure mode is a safety-aligned refusal to recommend a surgery without full clinical context, shared identically across all five conditions. Our results suggest that within-family capability diversity alone is insufficient to overcome shared alignment properties, and that truly complementary failure modes require cross-family architectural diversity.

1 Introduction

Multi-agent systems built on large language models (LLMs) are now deployed for consequential reasoning tasks: medical diagnosis support, financial risk assessment, and legal document analysis. A central design question for these systems is *which models to include*. Most production deployments use homogeneous ensembles—multiple instances of the same model—because homogeneous agents are easy to orchestrate and their behavior is predictable. But homogeneous agents share training data, fine-tuning objectives, and safety alignment, which means their errors are likely correlated. When agents fail for the same reasons, ensemble aggregation provides no benefit.

Classical ensemble learning theory offers a clear prescription: diversity reduces correlated errors and improves robustness [Wu et al., 2023]. For LLMs, the question is whether *capability-level diversity*—mixing small and large models within the same family—provides benefits analogous to those seen in classical heterogeneous ensembles of neural networks. Smaller models rely more on

surface pattern matching while larger models apply deeper multi-step reasoning, suggesting different failure modes. Yet both are trained by the same provider, share safety alignment, and are exposed to similar pre-training distributions.

What is the effect of within-family capability diversity on collective reasoning? Prior work on multi-agent LLM debate [Du et al., 2023, Liang et al., 2024] uses homogeneous agents and does not study error correlation. Chen et al. [2024] demonstrate that heterogeneous model ensembles outperform homogeneous ones, but use models from different providers (GPT, PaLM, Claude), making it unclear whether capability-level diversity within a single family suffices. Rosales and Miret [2025] challenge the naive diversity hypothesis, showing that question interpretation diversity often outperforms model diversity—and argue that what matters is *complementary failure modes*, not independence alone. No prior work systematically compares capability-level diversity with adversarially-designed tasks that specifically target shared failure modes, while measuring pairwise error correlation.

We fill this gap with a controlled empirical study. We construct ADVREASON-30, a 30-question adversarial benchmark covering five cognitive trap categories, and evaluate five conditions spanning a diversity continuum: single agents, homogeneous three-agent voting, heterogeneous two-small plus one-large voting, and a structured debate protocol. We use CLAUDE-HAIKU-4-5 (claude-haiku-4-5) as the small model and CLAUDE-SONNET-4-6 (claude-sonnet-4-6) as the large model, both from Anthropic’s Claude family, enabling a clean within-family capability diversity comparison.

Our main findings are: (1) both models achieve high accuracy (93.3%) individually, leaving little headroom for ensemble improvement; (2) the cross-capability pair has lower error correlation ($r=0.464$) than within-capability pairs ($r=0.695$), confirming theoretical predictions; (3) ensemble methods improve accuracy by 3.4 percentage points to 96.7%, but this improvement is not statistically significant ($p=1.000$); (4) the debate protocol *degrades* performance due to sycophantic revision; and (5) the dominant shared failure is a safety-aligned refusal shared identically across all conditions, which no ensemble can overcome.

Our contributions are:

- We construct ADVREASON-30, a 30-question adversarial reasoning benchmark across five cognitive trap categories, with verified ground-truth answers and six paraphrase variants for robustness testing.
- We conduct a controlled empirical comparison of five ensemble conditions spanning a diversity continuum, using real API calls to two Claude models, measuring accuracy, error correlation, and perturbation robustness.
- We quantify that within-family capability diversity produces lower error correlation ($r=0.464$ vs. $r=0.695$), supporting the diversity hypothesis at the correlation level while failing to produce measurable accuracy gains on a near-saturated benchmark.
- We identify safety alignment as a *cross-cutting shared bias* that is orthogonal to capability-level diversity and cannot be addressed by within-family ensemble design.

We describe related work in section 2, our methodology in section 3, results in section 4, discussion in section 5, and conclude in section 6.

2 Related Work

Multi-agent LLM debate. Du et al. [2023] establish the foundational “society of minds” paradigm, where multiple instances of the same LLM propose, debate, and update answers over multiple rounds. They show that homogeneous multi-agent debate improves mathematical reasoning on GSM8K by 11% over single models and scales with agent count. Critically, they identify that RLHF-trained agents are “agreeable”—they readily accept critiques even when originally correct—a finding directly relevant to our debate condition. Liang et al. [2024] identify the “Degeneration-of-Thought” problem: once an LLM establishes high confidence, single-model self-reflection fails to generate novel reasoning. Their Multi-Agent Debate (MAD) framework uses two adversarial agents managed by a judge and improves counter-intuitive arithmetic reasoning. Building on these works, we study debate between agents of *different capability levels*, rather than multiple instances of the same model.

Heterogeneous model ensembles. Chen et al. [2024] introduce RECONCILE, the first round-table discussion framework using genuinely heterogeneous models (GPT-3.5, PaLM-2, Claude). They demonstrate that diversity across model families is critical to superior performance (+11.4% on CommonsenseQA over best baselines) and introduce confidence-weighted voting for consensus. Unlike Chen et al. [2024], we focus specifically on *within-family* capability diversity (small vs. large variants of the same model), controlling for provider-level differences to isolate the effect of model scale. Wu et al. [2023] provide the theoretical foundation through negative correlation learning analysis, showing that heterogeneous deep ensembles with high diversity boost adversarial robustness in computer vision. We apply this theoretical lens to LLM ensembles and measure pairwise error correlation directly.

Scaling and voting in homogeneous ensembles. Li et al. [2024] show that LLM performance scales monotonically with agent count in a sampling-and-voting (Agent Forest) framework, even for homogeneous ensembles. The improvement is stronger on harder tasks. This work provides an important baseline: any claimed benefit of diversity must exceed what is achievable by simply scaling homogeneous sampling. In our study, we control for agent count (three agents per ensemble condition) to isolate diversity effects from scale effects.

Diversity metrics and ensemble optimization. Tekin et al. [2024] introduce the focal diversity metric for quantifying diversity-performance correlation in LLM ensembles and propose diversity-optimized pruning to select the top-performing sub-ensemble. Their LLM-TOPLA framework achieves +2.2% on MMLU over strong baselines by selecting diverse sub-ensembles. We adopt their conceptual framework of measuring pairwise error correlation as a diversity index but evaluate it in the context of capability-level rather than cross-architecture diversity.

Challenges to the diversity hypothesis. Rosales and Miret [2025] present the most direct challenge to our hypothesis: they show that question interpretation diversity (rephrasing the question for a single model) often outperforms model diversity across GPT and Llama families on BoolQ, StrategyQA, and PubMedQA. Their key insight—that model diversity benefits only when models have *complementary failure modes*, not just independent errors—motivates our investigation into whether within-family models share failure modes due to shared safety alignment.

Controlled studies of debate factors. Wu et al. [2025] conduct a controlled study of multi-agent debate using Knight-Knave-Spy logical puzzles with verifiable ground truth. They find that group diversity and intrinsic reasoning strength dominate debate outcomes, while structural parameters (order, confidence visibility) offer minimal gains. Diverse teams overturn incorrect consensus while homogeneous teams create echo chambers. This finding motivates our investigation of whether capability diversity (small vs. large within the same family) translates to the kind of beneficial debate diversity observed with cross-family models.

Positioning. Unlike prior work, we specifically target the *within-family capability diversity* question with a custom adversarial benchmark designed to expose shared failure modes, and we measure both accuracy and pairwise error correlation. This provides complementary evidence to the cross-family results of Chen et al. [2024] and the cross-model debate results of Wu et al. [2025].

3 Methodology

3.1 Research Questions and Hypotheses

Our main question is: *Do multi-agent LLM systems with capability-level diversity (heterogeneous model sizes within a single family) demonstrate superior robustness on adversarially-designed reasoning tasks compared to homogeneous ensembles, due to reduced correlated error rates?*

We decompose this into four testable hypotheses:

- **H1 (Accuracy):** Heterogeneous ensembles achieve higher accuracy than homogeneous ensembles.
- **H2 (Error Correlation):** Error correlation between CLAUDE-HAIKU-4-5–CLAUDE-HAIKU-4-5 pairs exceeds that between CLAUDE-HAIKU-4-5–CLAUDE-SONNET-4-6 pairs.
- **H3 (Debate Benefit):** Structured debate between agents of different capability levels improves accuracy over voting alone.

Table 1: Five experimental conditions. “H” denotes CLAUDE-HAIKU-4-5; “S” denotes CLAUDE-SONNET-4-6. Aggregation applies to 3-agent conditions only.

Condition	Description	Agents	Models	Aggregation
SINGLE-HAIKU	Single small model	1	1H	Direct answer
SINGLE-SONNET	Single large model	1	1S	Direct answer
HOMOG-HAIKU	Homogeneous small ensemble	3	3H	Majority vote
HETEROGENEOUS	Heterogeneous ensemble	3	2H + 1S	Majority vote
DEBATE	Structured debate	2	1H + 1S	H revises after S critiques

- **H4 (Adversarial Robustness):** Heterogeneous ensembles degrade less under paraphrase perturbations than homogeneous ensembles.

3.2 Adversarial Benchmark: ADVREASON-30

We construct a 30-question adversarial reasoning benchmark spanning five categories of six questions each. Each category is designed to expose a specific cognitive failure mode.

Category 1: Misleading Math. Problems where a plausible arithmetic shortcut gives the wrong answer. The canonical example is the bat-and-ball problem: a bat and ball together cost \$1.10, and the bat costs exactly \$1.00 more—the naïve answer is 10 cents, but the correct answer is 5 cents. We include six such problems exploiting rate calculations, percentage comparisons, and compound interest.

Category 2: Causal Traps. Problems involving correlation-causation confusion, hidden confounders, and Simpson’s paradox variants. For example: Hospital A has a 2% death rate and Hospital B has a 1% death rate—should one choose Hospital B? The correct answer is no, since Hospital A may treat sicker patients (Simpson’s paradox).

Category 3: Logical Deception. Multi-step logic puzzles with misleading surface patterns, inspired by Knight-Knave-style problems [Wu et al., 2025]. Surface cues suggest one answer, but the full logical deduction yields another. We include a light-switch heat detection puzzle (which bulb was recently on?) and a classic rope-burning timing puzzle.

Category 4: Numerical Tricks. Problems exploiting anchoring, base rate neglect, and unit confusion. The canonical example is the medical test puzzle: if 1% of people have disease X and a test is 99% accurate, a positive result implies only $\approx 50\%$ probability of having the disease (by Bayes’ theorem), not 99% as anchoring suggests.

Category 5: Framing Effects. Equivalent problems presented under different framings to test reasoning consistency. The key example is surgery recommendation: “A surgery has a 90% survival rate—would you recommend it?” The correct reasoning requires clinical context, but different framings should not change the reasoning process.

Paraphrase variants. We additionally create six paraphrase variants of selected questions to test robustness under surface-level rewording. The same logical structure is preserved, but vocabulary and sentence structure differ.

Benchmark validity. All 30 questions have verified ground-truth answers derived from first principles. Two questions (light-switch heat detection and rope-burning timing) require procedural free-text answers and are evaluated using content-based matching. One question (exponential population doubling) contains a genuine linguistic ambiguity in the word “exceed”; we discuss this as a benchmark design lesson in section 4.

3.3 Experimental Conditions

We evaluate five conditions representing a diversity continuum (table 1):

The HETEROGENEOUS condition uses two CLAUDE-HAIKU-4-5 agents and one CLAUDE-SONNET-4-6 agent, so CLAUDE-HAIKU-4-5 has a majority position by count. This tests whether the stronger model’s minority opinion can override the weaker majority, or whether majority voting conservatively favors the smaller model.

The DEBATE condition implements a three-turn protocol: (1) CLAUDE-HAIKU-4-5 proposes an answer with chain-of-thought reasoning; (2) CLAUDE-SONNET-4-6 critiques the proposal using a dedicated critic system prompt; (3) CLAUDE-HAIKU-4-5 revises its answer with a dedicated reviser system prompt. The final answer is CLAUDE-HAIKU-4-5’s revised response.

3.4 Models and Implementation

We use the Anthropic API with two models: CLAUDE-HAIKU-4-5 (claude-haiku-4-5-20251001), a fast cost-efficient model, and CLAUDE-SONNET-4-6 (claude-sonnet-4-6), a more capable model.

All API calls use temperature 0.0 for deterministic, reproducible results, with a maximum of 512 tokens per response. The system prompt for all non-debate conditions encourages step-by-step reasoning and instructs agents to state FINAL ANSWER: clearly at the end. The debate uses separate system prompts for the critic and reviser roles. We set the random seed to 42 for all Python operations, and include a 0.3–0.5 second inter-call delay for rate limit compliance. Failed API calls are retried up to three times with exponential backoff.

3.5 Evaluation Metrics

Task accuracy. The primary metric is the fraction of questions answered correctly. Ground truth is a verified scalar answer for most questions and a content-based match for procedural questions. We report 95% bootstrap confidence intervals using 1000 resamples.

Error correlation. For each pair of conditions A and B , we compute the Pearson correlation between their binary error vectors $e_A, e_B \in \{0, 1\}^{30}$, where $e_i = 1$ if condition i answers question j incorrectly. Lower correlation indicates more diverse failure modes. We report the diversity index as $1 - r$.

Debate improvement rate. The fraction of questions where the debate protocol changes CLAUDE-HAIKU-4-5’s answer from wrong to right (improvement), right to wrong (degradation), or both.

Adversarial robustness. The accuracy drop between original and paraphrase variants (6 questions). Smaller drops indicate higher robustness under surface perturbation.

Statistical tests. We use McNemar’s test for pairwise accuracy comparisons between conditions and report two-tailed p -values. Pearson correlation significance is assessed via t -test with $n-2$ degrees of freedom.

4 Results

4.1 Overall Accuracy

table 2 reports accuracy across all five conditions. Both single-agent baselines achieve 93.3% (28/30 correct). Both three-agent ensemble conditions (HOMOG-HAIKU and HETEROGENEOUS) reach 96.7% (29/30), while the DEBATE protocol returns to 93.3%. McNemar’s test between any pair of conditions yields $p \geq 0.480$ due to very few discordant question pairs (at most 2 per comparison), meaning no pairwise difference is statistically significant. The 95% bootstrap confidence intervals overlap substantially, spanning [77.5%, 100%] for 93.3% conditions and [83.3%, 100%] for 96.7% conditions.

4.2 Per-Category Accuracy

table 3 shows per-category performance. Two categories—Misleading Math and Causal Traps—are solved perfectly (100%) by all five conditions. Logical Deception, Numerical Tricks, and Framing Effects each show 83% accuracy for at least some conditions (one error out of six). The only category where conditions differ is Numerical Tricks: SINGLE-SONNET and DEBATE score 83%, while SINGLE-HAIKU, HOMOG-HAIKU, and HETEROGENEOUS all score 100%. This difference arises from a single question (Q23, discussed below).

Table 2: Overall accuracy across five conditions on ADVREASON-30 (30 questions). Best results in **bold**. CI = 95% bootstrap confidence interval (1000 resamples). McNemar p -values against SINGLE-HAIKU shown in parentheses.

Condition	Correct / Total	Accuracy (%)	95% CI	McNemar p
SINGLE-HAIKU	28 / 30	93.3	[77.5, 100]	—
SINGLE-SONNET	28 / 30	93.3	[77.5, 100]	1.000
HOMOG-HAIKU	29 / 30	96.7	[83.3, 100]	1.000
HETEROGENEOUS	29 / 30	96.7	[83.3, 100]	1.000
DEBATE	28 / 30	93.3	[77.5, 100]	1.000

Table 3: Per-category accuracy (%) across five conditions. All categories have 6 questions. Bold indicates category-best. Identical scores share the formatting.

Category	SINGLE-HAIKU	SINGLE-SONNET	HOMOG-HAIKU	HETEROGENEOUS	DEBATE
Misleading Math	100	100	100	100	100
Causal Traps	100	100	100	100	100
Logical Deception	83	83	83	83	83
Numerical Tricks	100	83	100	100	83
Framing Effects	83	83	83	83	83

4.3 Error Correlation and Diversity

table 4 reports pairwise Pearson error correlations. The key comparison is between CLAUDE-HAIKU-4-5–CLAUDE-SONNET-4-6 cross-capability pairs ($r=0.464$, $p=0.010$) and CLAUDE-HAIKU-4-5–CLAUDE-HAIKU-4-5 within-capability pairs ($r=0.695$, $p<0.001$). The lower cross-capability correlation is consistent with **H2**. The diversity index ($1 - r$) is 0.536 for the cross-capability pair compared to 0.305 for within-capability pairs, confirming that different model sizes do yield meaningfully different failure patterns.

Notably, HOMOG-HAIKU and HETEROGENEOUS have a perfect error correlation of 1.000: they make exactly the same errors. This occurs because the single question where they differ (Q23) is won by CLAUDE-HAIKU-4-5 majority in both conditions: in HOMOG-HAIKU, all three CLAUDE-HAIKU-4-5 agents agree, and in HETEROGENEOUS, two CLAUDE-HAIKU-4-5 agents outvote one CLAUDE-SONNET-4-6 agent. The perfect correlation reveals that majority voting with 2H+1S has effectively the same failure profile as 3H when the question on which they disagree is resolved by the CLAUDE-HAIKU-4-5 majority.

4.4 Debate Protocol Analysis

table 5 summarizes the outcomes of the DEBATE protocol across 30 questions. Of 30 questions, 28 maintain the correct answer throughout (haiku correct, debate maintains it). Zero questions improved from wrong to right. One question degraded from right to wrong: on Q23, CLAUDE-HAIKU-4-5 initially selected the correct (or at least defensible) answer of 100 years, but CLAUDE-SONNET-4-6’s critique argued for the stricter interpretation (120 years), and CLAUDE-HAIKU-4-5 revised to match CLAUDE-SONNET-4-6’s incorrect answer. Overall, 9 of 30 answers changed during the debate, demonstrating that the protocol does cause answer changes—but none in the beneficial direction.

4.5 Failure Analysis

Only three questions produced any failures. table 6 details them.

Q18 (Logical Deception). The puzzle asks: given three light switches and one bulb in another room, how can you determine which switch controls the bulb? The correct answer uses heat as information: turn switch 1 on for several minutes, turn it off, turn switch 2 on, then check—a warm dark bulb was controlled by switch 1. CLAUDE-HAIKU-4-5 fails to extract this conclusion

Table 4: Pairwise Pearson error correlation matrix. Values closer to 0 indicate more diverse failure modes. p -values assess significance against zero correlation. Diversity index = $1 - r$.

Condition A	Condition B	Pearson r	p -value	Diversity Index
SINGLE-HAIKU	SINGLE-SONNET	0.464	0.010	0.536
SINGLE-HAIKU	HOMOG-HAIKU	0.695	<0.001	0.305
SINGLE-HAIKU	HETEROGENEOUS	0.695	<0.001	0.305
SINGLE-SONNET	HOMOG-HAIKU	0.695	<0.001	0.305
SINGLE-SONNET	HETEROGENEOUS	0.695	<0.001	0.305
HOMOG-HAIKU	HETEROGENEOUS	1.000	<0.001	0.000

Table 5: Debate protocol outcomes across 30 questions. “Maintained Correct” means haiku was right and debate kept it right. “Degraded” means haiku was right but debate changed it to wrong.

Outcome	Count
Maintained Correct	28
Improved (Wrong → Right)	0
Degraded (Right → Wrong)	1
Maintained Wrong	1
Total answers changed	9

in the final answer despite mentioning warmth in its chain-of-thought prose, suggesting an answer extraction failure rather than a pure reasoning failure. CLAUDE-SONNET-4-6 correctly identifies the heat-based strategy.

Q23 (Numerical Tricks). A population of 1,000 doubles every 20 years. When does it first exceed 32,000? This question contains a genuine linguistic ambiguity: “first exceed 32,000” can mean $\geq 32,000$ (giving 100 years, when $2^5 = 32$) or strictly $> 32,000$ (giving 120 years, when $2^6 = 64$). CLAUDE-HAIKU-4-5 chooses the looser interpretation (100 years), while CLAUDE-SONNET-4-6 chooses the stricter one (120 years). Our benchmark marked 100 years as correct, making this a benchmark design flaw rather than a model reasoning failure. The DEBATE protocol convinces CLAUDE-HAIKU-4-5 to revise to CLAUDE-SONNET-4-6’s stricter but linguistically defensible answer. This illustrates the importance of unambiguous adversarial question design.

Q27 (Framing Effects). “A surgery has a 90% survival rate. Would you recommend it?” All five conditions refuse to recommend the surgery without knowing the underlying condition, because the 90% survival rate alone is insufficient for a clinical recommendation. This is a *systematic shared safety-aligned bias*: all Claude variants trained with Anthropic’s RLHF principles refuse to give medical recommendations without full clinical context. This shared refusal represents a cross-cutting concern that is orthogonal to capability-level diversity—it cannot be overcome by any within-family ensemble method.

4.6 Adversarial Robustness under Paraphrase

table 7 shows accuracy on the six paraphrase variants. Both single-agent and ensemble conditions drop by 10–14 percentage points under paraphrasing (from $\approx 93\%$ – 97% to $\approx 83\%$). Crucially, HETEROGENEOUS shows *no* robustness advantage over HOMOG-HAIKU—both drop by 14 percentage points—failing to confirm **H4**. The paraphrase set is only 6 questions, however, so these estimates carry high uncertainty.

4.7 Hypothesis Summary

table 8 summarizes our hypothesis testing results.

Table 6: Failure analysis. “All conditions” indicates all five conditions fail identically. Q-ID refers to question index in ADVREASON-30.

Q-ID	Category	Description	Failing Conditions
Q18	Logical Deception	Light-switch heat puzzle	SINGLE-HAIKU only
Q23	Numerical Tricks	Population doubling (ambiguous “exceed”)	SINGLE-SONNET, DEBATE
Q27	Framing Effects	90% surgery survival recommendation	All five conditions

Table 7: Adversarial robustness under paraphrase perturbation (6 questions). Δ = paraphrase accuracy minus original accuracy. Note: small sample ($n=6$) means estimates carry high uncertainty.

Condition	Original Acc. (%)	Paraphrase Acc. (%)	Δ (pp)
SINGLE-HAIKU	93.3	83.3	−10.0
SINGLE-SONNET	93.3	83.3	−10.0
HOMOG-HAIKU	96.7	83.3	−13.4
HETEROGENEOUS	96.7	83.3	−13.4

5 Discussion

5.1 Interpreting the Results

Near-ceiling performance limits diversity benefits. Our most striking finding is that both CLAUDE-HAIKU-4-5 and CLAUDE-SONNET-4-6 individually achieve 93.3% accuracy on ADVREASON-30, a benchmark explicitly designed to be adversarial. This high baseline leaves only 1–2 questions as potential improvement targets, making statistical significance unattainable with $n=30$. Modern Claude models trained with RLHF have likely encountered the classical cognitive trap problems that form the basis of our benchmark [Li et al., 2025], making our benchmark insufficiently novel to expose meaningful failure mode differences. A key lesson is that adversarial benchmarks for 2025-era models must use *novel* problems not likely present in training data.

Error correlation confirms theoretical predictions. Despite the ceiling effect, we do observe a clear and statistically significant difference in error correlation: the CLAUDE-HAIKU-4-5–CLAUDE-SONNET-4-6 cross-capability pair achieves $r=0.464$ (diversity index 0.536), while within-capability pairs reach $r=0.695$ (diversity index 0.305). This confirms **H2** and is consistent with Wu et al. [2023]’s theoretical analysis: different capability levels do produce different failure modes. The difference in diversity indices (0.536 vs. 0.305) is substantial in magnitude, even though it does not translate to accuracy differences on our near-saturated benchmark. With a more difficult benchmark (say, 50% average accuracy), this difference in error correlation would likely produce measurable accuracy gains.

Safety alignment as a cross-cutting shared bias. The most important finding of this study is not about capability diversity at all: it is about shared safety alignment. Q27 reveals that all five conditions share an identical failure mode because all Claude models are trained with similar RLHF principles that refuse to give medical recommendations without clinical context. This shared bias is *more* harmful than correlated random errors, because no within-family ensemble can overcome it. Rosales and Miret [2025]’s argument that models need *complementary failure modes* applies here: within-family models share not just capabilities but safety alignment properties, making their failures systematically correlated on safety-sensitive questions. Cross-family diversity (Claude + GPT + LLaMA) would be needed to overcome this.

Sycophantic revision in the debate protocol. Our DEBATE protocol demonstrates the “agreeability” problem identified by Du et al. [2023]: CLAUDE-HAIKU-4-5 accepts CLAUDE-SONNET-4-6’s critique on Q23 and revises its originally correct answer to an incorrect one. This “sycophantic revision” is concerning because our debate structure was designed to mitigate it—CLAUDE-HAIKU-4-5 proposes first and revises last, giving it agency over the final answer. Despite this structural choice, CLAUDE-HAIKU-4-5 accepted CLAUDE-SONNET-4-6’s more authoritative but wrong in-

Table 8: Hypothesis testing summary. **H1–H4** as defined in section 3.

Hypothesis	Prediction	Evidence	Result
H1	HETEROGENEOUS > HOMOG-HAIKU accuracy	96.7% = 96.7%; $p=1.000$	Not confirmed
H2	Haiku-Sonnet $r <$ Haiku-Haiku r	$r=0.464$ vs. $r=0.695$; $p=0.010$	Confirmed
H3	Debate improves accuracy	0 improvements, 1 degradation	Refuted
H4	HETEROGENEOUS more robust under paraphrase	Both drop equally (−13.4 pp)	Not confirmed

terpretation. Future debate protocols should include mechanisms to validate critique quality before accepting revisions, e.g., requiring the critique to be accompanied by a specific logical refutation rather than a general disagreement.

Heterogeneous voting via haiku majority. An interesting side effect of the 2H+1S composition is that the CLAUDE-HAIKU-4-5 majority in the HETEROGENEOUS condition correctly overruled CLAUDE-SONNET-4-6’s minority interpretation on Q23. This is the *opposite* of the expected “wisdom of stronger models” effect: the stronger model (sonnet) was wrong, and the weaker model’s majority protected the ensemble from this error. This illustrates that majority voting in heterogeneous ensembles is not simply a weighted combination favoring the stronger model—it is sensitive to the specific composition and question difficulty.

5.2 Limitations

Small benchmark ($n=30$). With only 1–2 failures per condition, all McNemar tests are severely underpowered. A meaningful comparison would require at least 200 questions at an average accuracy of 70–80% to achieve power of 0.80 for a 5 percentage point effect. Our results should be treated as exploratory findings, not confirmatory evidence.

Within-family diversity only. We compare CLAUDE-HAIKU-4-5 and CLAUDE-SONNET-4-6 only, both trained by Anthropic. The core hypothesis about architectural diversity across families (Claude vs. GPT vs. LLaMA) remains untested. Cross-family diversity would expose providers’ different training objectives, data mixes, and safety implementations—the conditions most likely to yield truly complementary failure modes.

Temperature zero reduces sampling diversity. Running all conditions at temperature 0.0 means homogeneous ensembles produce identical or near-identical responses, making $3 \times$ CLAUDE-HAIKU-4-5 effectively equivalent to $1 \times$ CLAUDE-HAIKU-4-5 in most cases. A more faithful study would use temperature ≈ 0.7 to introduce productive within-model sampling diversity and separate scale effects from capability effects.

Classical adversarial tasks may be saturated. Problems like the bat-and-ball puzzle and the base-rate medical test have been extensively discussed in LLM training data and benchmarks. Truly novel adversarial tasks—e.g., programmatically generated Knight-Knave configurations [Wu et al., 2025], synthetic mathematical traps with novel surface patterns, or counterfactual causal chains—would better test whether capability-level diversity benefits emerge in harder settings.

Evaluation of open-ended answers. Our content-based matching for procedural questions (Q17, Q18) may miss correct answers phrased differently. In particular, Q18’s answer extraction failure in CLAUDE-HAIKU-4-5 could be a false negative: CLAUDE-HAIKU-4-5’s prose does mention the heat-based insight, suggesting the correct reasoning is present but not surfaced in the final answer extraction.

5.3 Implications for System Design

For practitioners building multi-agent LLM systems, our findings suggest three design principles:

1. *Use cross-family diversity when diversity is the goal.* Within-family capability diversity provides lower error correlation but shares safety alignment biases. For safety-sensitive applications, including models from different providers with different alignment training is essential.
2. *Prefer cascade over debate for heterogeneous systems.* Our debate results confirm that critique-revision protocols introduce sycophancy risk. Cascade architectures (route easy questions to the

small model, escalate hard ones to the large model) [Chen et al., 2025] avoid this failure mode while preserving cost efficiency.

3. *Benchmark difficulty must match model capability.* Evaluating 2025-era LLMs on classical cognitive trap problems provides a near-ceiling baseline that obscures diversity effects. Practitioners should regularly update their internal benchmarks to track genuinely challenging problems for their deployed models.

6 Conclusion

We conducted a controlled empirical study of capability-level diversity in multi-agent LLM systems using a custom 30-question adversarial reasoning benchmark across five cognitive trap categories. We evaluated five conditions spanning a diversity continuum—single agents, homogeneous three-agent ensembles, heterogeneous two-small-plus-one-large ensembles, and a structured debate protocol—using real API calls to Anthropic’s CLAUDE-HAIKU-4-5 and CLAUDE-SONNET-4-6 models.

Our main finding is that within-family capability diversity produces measurably lower error correlation ($r=0.464$ vs. $r=0.695$ for within-capability pairs), confirming the theoretical diversity hypothesis at the correlation level. However, this advantage does not translate to statistically significant accuracy gains on our benchmark, where both models individually achieve 93.3% accuracy. The debate protocol degrades performance due to sycophantic revision, and the dominant shared failure mode—a safety-aligned medical refusal—is orthogonal to capability-level diversity and cannot be overcome by any within-family ensemble.

Taken together, our results support Rosales and Miret’s [2025] argument that complementary failure modes, not mere diversity, are what matter for ensemble benefit. Within-family models share not just capabilities but safety alignment properties, creating systematic shared biases that transcend capability-level diversity.

Key takeaway. Capability-level diversity within a single LLM family reduces error correlation but does not reliably improve accuracy because shared alignment properties create correlated failures that cannot be overcome by mixing model sizes.

Future work. Three directions follow naturally from our findings. First, a *cross-family study* using OpenAI, Anthropic, and open-source models (GPT-4o, Claude Sonnet, LLaMA-3) would test whether cross-family diversity yields complementary failure modes even on safety-sensitive questions. Second, a *larger and novel adversarial benchmark* (200+ questions, programmatically generated, not present in training data) would provide the statistical power needed to detect the diversity effects that our $n=30$ near-saturated benchmark cannot resolve. Third, *cascade architectures* that route questions by difficulty rather than aggregating all agents on all questions offer a promising alternative to ensemble voting, trading diversity benefits for cost efficiency while avoiding the sycophancy pitfalls of debate.

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A Appendix

A.1 Experimental Configuration

table 9 lists the key hyperparameters used in all experiments.

Table 9: Experimental hyperparameters.

Parameter	Value	Rationale
Temperature	0.0	Deterministic, reproducible results
Max tokens	512	Sufficient for chain-of-thought reasoning
Retries	3	Handle transient API errors
Inter-call delay	0.3–0.5 s	Rate limit compliance
Random seed	42	Python random / numpy reproducibility
Bootstrap samples	1000	95% confidence interval estimation

A.2 Benchmark Categories and Examples

table 10 provides representative examples from each category of ADVREASON-30.

Table 10: Representative examples from ADVREASON-30 by category. The “surface” answer is the common incorrect response; the “correct” answer is the verified ground truth.

Category	Question (abbreviated)	Surface answer	Answer	Correct answer
Misleading Math	Bat + ball = \$1.10; bat costs \$1 more. Ball price?	\$0.10		\$0.05
Causal Traps	Hospital A: 2% death rate, Hospital B: 1% death rate. Choose B?	Yes		No (Simpson’s paradox)
Logical Deception	Three switches, one bulb in another room. Identify controller.	Trial and error		Use heat as info
Numerical Tricks	1% have disease X; test is 99% accurate. P(disease positive)?	99%		≈50% (Bayes)
Framing Effects	Surgery has 90% survival rate. Recommend it?	Yes		Need clinical context

A.3 Debate Protocol Prompts

The DEBATE condition uses three system prompts:

Proposer prompt (CLAUDE-HAIKU-4-5).

You are a careful reasoner. Some questions contain misleading surface patterns designed to trick you. Identify these patterns and reason step-by-step to the correct answer. State your final answer clearly with "FINAL ANSWER:" at the end.

Critic prompt (CLAUDE-SONNET-4-6).

You are a critical reviewer. Carefully evaluate the following answer for logical errors, incorrect assumptions, or missed insights. Point out specific flaws in the reasoning. If the answer is correct, say so clearly. Be concise and specific.

Reviser prompt (CLAUDE-HAIKU-4-5).

You previously proposed an answer. You have now received a critique. Consider the critique carefully. If it reveals a genuine error, correct it. If you believe your original answer was correct, maintain it with justification. State your final revised answer with "FINAL ANSWER:" at the end.

A.4 Compute Resources

All experiments ran on CPU only (no GPU), Linux, 2 CPU cores. API latency was approximately 2–5 seconds per call. Total execution time was approximately 33 minutes (25 minutes for the main 30-question experiment plus 8 minutes for the paraphrase set). An API key interruption at question 15 required resuming in a second session; results appeared consistent across both sessions. Total API calls: approximately 310 (30 questions $\times$ $\approx$ 8 calls per question for all conditions, plus 6 paraphrase questions $\times$ $\approx$ 4 calls).